Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** GLUE* | ***Context:** Luxembourgish Public Administration Civil Servants — GLUE Assessment*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GLUE's output ontology is hard single-label classification (binary or three-class) plus one regression task [Q12], scored via accuracy/MCC/F1/Pearson [Q20, Q23, Q30, Q33, Q35] aggregated as a macro-average [Q59, Q60]. The deployment requires multi-label classification (a single citizen message may concern multiple departments simultaneously) plus confidence scoring to flag uncertain cases for human review (elicitation A4). The dataset analysis confirms 'every classification task in the sampled data uses exactly one label per example' and 'no example has multiple labels, soft labels, or confidence scores.' Additionally, the output categories themselves (grammatical/ungrammatical, positive/negative, entailment/neutral/contradiction) have no correspondence to administrative routing categories: no État-vs-commune label, no frontalier-vs-resident label, no department-routing label, no urgency flag. OO is HIGH priority.

ID	Evidence
Q12	“All tasks are single sentence or sentence pair classification, except STS-B, which is a regression task. MNLI has three classes; all other classification tasks have two. Test sets shown in bold use labels that have never been made public in any form.” (p.2)
Q20	“Unless otherwise mentioned, tasks are evaluated on accuracy and are balanced across classes.” (p.3)
Q23	“Following the authors, we use Matthews correlation coefficient (Matthews, 1975) as the evaluation metric, which evaluates performance on unbalanced binary classification and ranges from -1 to 1, with 0 being the performance of uninformed guessing.” (p.3)
Q30	“Because the classes are imbalanced (68% positive), we follow common practice and report both accuracy and F1 score.” (p.3)
Q33	“As in MRPC, the class distribution in QQP is unbalanced (63% negative), so we report both accuracy and F1 score.” (p.3)
Q35	“The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients.” (p.3)
Q59	“The benchmark site shows per-task scores and a macro-average of those scores to determine a system's position on the leaderboard.” (p.5)
Q60	“For tasks with multiple metrics (e.g., accuracy and F1), we use an unweighted average of the metrics as the score for the task when computing the overall macro-average.” (p.5)

Strengths

- GLUE explicitly diversifies metrics across tasks — using MCC for unbalanced binary classification [Q23] and Pearson/Spearman for regression [Q35] — modeling awareness that a single accuracy metric cannot capture all evaluation needs. This metric-diversity principle is transferable to a multi-label, confidence-scored Luxembourgish benchmark.

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Q23	“Following the authors, we use Matthews correlation coefficient (Matthews, 1975) as the evaluation metric, which evaluates performance on unbalanced binary classification and ranges from -1 to 1, with 0 being the performance of uninformed guessing.” (p.3)
Q35	“The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients.” (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output categories are not regionally relevant. CoLA's grammatical/ungrammatical [Q22], SST-2's positive/negative [Q28], MNLI's entailment/neutral/contradiction [Q56], and RTE's entailment/not-entailment [Q49] do not map to any required deployment output: department routing, urgency flag, frustration flag, frontalier classification, État/commune competency.

OO-2: Missing categories specific to the regional context include: department-level routing labels across both État ministries and 100 communes [WEB-16, WEB-17], frontalier-vs-resident classification [WEB-6, WEB-8], urgency level (housing political-sensitivity flag, social-security urgency), and intent classes for administrative correspondence (request, complaint, status inquiry, document submission). None exist in GLUE.

OO-3: GLUE's sentiment ontology (positive/negative [Q28]) encodes English movie-review evaluative norms; the deployment requires a frustration/urgency ontology calibrated to understated continental-European formal administrative register, where direct expressions of frustration are rare (cultural_norms_notes). The two label spaces are not compatible.

OO-4: Stakeholder-driven taxonomy redesign is not optional but required — GLUE's label spaces cannot be repurposed; a Luxembourg-specific output ontology must be constructed from scratch with civil-servant input. ltzGLUE [WEB-10] provides a closer Luxembourgish-language template (NER, intent, sentiment) but its taxonomies are still non-administrative.

OO-5: Structural-validity violation is fundamental: the structure of GLUE's output space (single hard labels, no multi-label, no confidence) misrepresents the routing decision structure the deployment requires. Content-validity violation is total: required categories are absent.

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Q12	“All tasks are single sentence or sentence pair classification, except STS-B, which is a regression task. MNLI has three classes; all other classification tasks have two. Test sets shown in bold use labels that have never been made public in any form.” (p.2)
Q22	“Each example is a sequence of words annotated with whether it is a grammatical English sentence.” (p.3)
Q28	“We use the two-way (positive/negative) class split, and use only sentence-level labels.” (p.3)
Q49	“We convert all datasets to a two-class split, where for three-class datasets we collapse neutral and contradiction into not entailment, for consistency.” (p.4)
Q56	“Labels are entailment (E), neutral (N), or contradiction (C).” (p.5)
WEB-6	~47% of workforce are frontaliers — establishes the frontalier-vs-resident label requirement (https://statistiques.public.lu/en/publications/series/regards/2025/regards-01-25.html)
WEB-8	https://www.leretourdelautruche.com/en/scale-up-en/managing-cross-border-employees-france-luxembourg-complete-hr-guide-2026/
WEB-10	ltzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)
WEB-16	100 communes as of 1 September 2023 — establishes communal-routing label space requirement absent from GLUE (https://elections.public.lu/en/fusion-communes.html)
WEB-17	https://gouvernement.lu/en/actualites/toutes_actualites/communiques/2023/02-fevrier/10-100-communes.html

Information Gaps

- The exact required output schema (number of routing labels, urgency-flag thresholds, multi-label co-occurrence patterns) is not specified in the elicitation beyond the multi-label-plus-confidence requirement.

Requires Expert Verification

- Civil-servant stakeholders should define the final routing taxonomy and confidence-threshold rules, including which uncertain cases are escalated to human review.

Remediation

Gap 1: GLUE's hard single-label output cannot represent the deployment's required multi-label routing with confidence scoring.

Recommendation: Design the evaluation output schema as multi-label classification with per-label confidence scores, plus separate threshold-tuning rules for human-in-the-loop escalation. Adapt GLUE's per-task metric diversity principle [Q23, Q35] by reporting micro/macro F1, label-wise precision/recall, and calibration metrics (e.g., expected calibration error) on the confidence outputs.

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